

Evaluating the Impact of Robot Carrying Capacity on Deadline-Constrained Warehouse Planning

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Abstract—Robot carrying capacity can influence both the physical execution of warehouse tasks and the computational difficulty of planning them. This work evaluates the effect of increasing a warehouse robot’s carrying capacity from one to two packages in a deadline-constrained PDDL+ planning domain. The original single-package model was generalized using numeric fluents for current load and maximum capacity. A controlled factorial experiment varied package count, deadline tightness, and package distribution while evaluating both capacity configurations on identical warehouse scenarios. Twenty reproducible seeds were generated for each condition, producing 160 paired scenarios and 320 ENHSP planner executions. Capacity one solved 72.50% of the scenarios, compared with 93.75% for capacity two. An exact McNemar test showed a significant paired difference ($p = 1.16 \times 10^{-10}$). Among the 116 scenarios solved by both capacities, capacity two reduced mean makespan by 28.54%, plan length by 23.34%, expanded nodes by 61.93%, and evaluated states by 57.80%. The results also show that the benefit of additional capacity depends strongly on package layout.

Index Terms—automated planning, PDDL+, warehouse robotics, numeric planning, carrying capacity, ENHSP

I. INTRODUCTION

Robotized warehouse systems introduce planning problems involving routing, order handling, task allocation, and autonomous vehicle operation [5], [6]. Large-scale robotic fulfillment systems have also demonstrated the practical need to coordinate fleets of autonomous vehicles in warehouse environments [12], while broader warehousing research highlights increasingly time-critical and automated fulfillment operations [13]. One robot characteristic that can directly affect these operations is carrying capacity.

A robot restricted to one package may need to revisit the same warehouse area several times, whereas a robot with greater capacity may collect multiple packages during a single visit. However, the benefit is not necessarily universal. If packages are spatially separated, additional capacity may not reduce travel. Under tight delivery deadlines, capacity may also determine whether a feasible plan exists.

Warehouse transportation tasks are related to classical pickup-and-delivery problems, where routing decisions interact with vehicle capacity, precedence, and temporal constraints [7], [8]. In this work, the problem is instead studied from an automated-planning perspective using an explicit symbolic and numeric model.

PDDL2.1 extended classical planning with durative actions and numeric fluents for temporal planning [9]. PDDL+ subsequently introduced autonomous processes and events, enabling

the representation of mixed discrete-continuous systems [1]. Planning such domains requires search methods that reason about numeric evolution, processes, and event timing [10], [11]. Numeric heuristic-search techniques further extend the range of expressive planning problems that can be addressed [2], [3], [4].

The research question addressed in this work is:

How does increasing robot carrying capacity from one to two packages affect solvability, mission makespan, plan length, and planning complexity under different package distributions and deadline constraints?

The main contributions are:

- a generalized capacity-aware PDDL+ warehouse model;
- a controlled paired experiment over 160 warehouse scenarios;
- automated ENHSP execution and metric extraction;
- statistical analysis of feasibility and planning efficiency;
- a ROS2 and Jupyter interface for interactive reproduction.

II. RELATED WORK

A. PDDL+ and Numeric Planning

Fox and Long first extended PDDL with durative actions and numeric fluents in PDDL2.1 [9]. PDDL+ subsequently added autonomous processes and events, allowing deterministic mixed discrete-continuous domains to be modeled explicitly [1]. This representation is suitable for warehouse tasks in which robot motion and deadline countdowns evolve continuously while pickups and deliveries remain discrete actions.

Coles and Coles studied planning with PDDL+ events and linear processes [10], while Piotrowski et al. investigated heuristic planning methods for PDDL+ domains [11]. These works illustrate the additional search challenges introduced by autonomous numeric evolution and event timing.

Scala, Haslum, and Thiébaux extended heuristic subgoalting techniques to numeric planning, including admissible heuristics for optimal numeric search [2]. Interval-based relaxations were subsequently proposed for more general numeric planning problems [3], while later work further developed subgoalting methods for satisficing and optimal numeric planning [4]. Together, these methods provide the planning background for the ENHSP-based experimental framework used here.

B. Robotic Warehouses and Pickup-and-Delivery

Azadeh et al. survey robotized and automated warehouse systems and identify routing, order picking, assignment, and system control as major research problems [5]. Barros and Nascimento provide a complementary survey focused on robotic mobile fulfillment systems [6]. Wurman et al. describe the coordination of large fleets of cooperative autonomous vehicles in the Kiva warehouse system [12], demonstrating the practical relevance of large-scale robotic material handling. Boysen et al. review warehousing in the e-commerce era and discuss the growing importance of automation, rapid order fulfillment, and new operational planning challenges [13].

The transportation component is closely related to pickup-and-delivery routing. Dumas et al. study pickup-and-delivery with time windows, combining routing, capacity, precedence, and temporal constraints [7]. Savelsbergh and Sol provide a broader formulation of the general pickup-and-delivery problem [8].

The present work differs by treating carrying capacity as an experimental variable inside a PDDL+ planning model and evaluating its influence not only on mission efficiency but also on planner solvability and search complexity.

III. METHODOLOGY

A. Warehouse Model

The PDDL+ domain contains robots, packages, and locations. The fixed warehouse topology includes a dock, two aisles, and a shipping location.

Robot movement is represented using the actions `start-move` and `end-move`, together with the continuous `robot-transit` process. The remaining travel distance decreases continuously while the robot is moving.

Each package has a numeric fluent `time-remaining`. For undelivered packages, the autonomous package-aging process decreases this value at unit rate:

$$\frac{d}{dt}\text{timeRemaining}(p) = -1. \quad (1)$$

When the remaining time reaches zero, a `deadline-breach` event marks the package as late. Delivery is permitted only while the package deadline has not been missed.

B. Generalized Carrying Capacity

The original model used the Boolean predicate `hand-empty`, enforcing a maximum capacity of one package.

The generalized model replaces it with:

- `current-load(robot)`;
- `max-capacity(robot)`.

A pickup is permitted when

$$\text{currentLoad}(r) < \text{maxCapacity}(r). \quad (2)$$

Pickup increments the current load by one, while intermediate drop and successful delivery decrement it by one. The same domain can therefore represent both capacity-one

TABLE I
GENERATED DEADLINE RANGES

Packages	Relaxed	Tight
2	20–30	7–12
3	25–40	11–16

and capacity-two robots by changing only the initial value of `max-capacity`.

This preserves the warehouse topology, package definitions, deadlines, actions, and planner configuration across both experimental conditions.

C. Package Handling and Planning Goal

Three discrete actions control package manipulation: `pick`, `drop-intermediate`, and `deliver-package`. A pickup marks a package as carried and increments the robot load. An intermediate drop places the package at the robot's current location and decrements the load. A delivery is valid only when the robot is carrying the package at its target location and the package deadline has not been violated.

For a capacity-two robot, the pickup precondition can remain satisfied after the first pickup. This allows two packages to be transported simultaneously when their locations and route structure make this useful.

The goal of every generated problem is the successful delivery of all packages. Consequently, the planner must jointly reason about movement, package manipulation, carrying capacity, and the continuous countdown of delivery deadlines.

IV. EXPERIMENTAL SETUP

Four factors are considered:

- carrying capacity: 1 or 2;
- package count: 2 or 3;
- deadline mode: relaxed or tight;
- package layout: clustered or distributed.

In clustered scenarios, packages occupy the same aisle. In distributed two-package scenarios, one package is located in each aisle. For three-package distributed scenarios, two packages share one aisle and the remaining package is placed in the other.

The deadline ranges used by the generator are shown in Table I. Tight ranges intentionally produce scenarios in which repeated travel can make timely delivery impossible.

The warehouse travel distances are fixed: dock–aisle A and aisle A–shipping require 2 time units, while dock–aisle B and aisle B–shipping require 3.

Twenty reproducible seeds are generated for each combination of package count, deadline mode, and layout:

$$2 \times 2 \times 2 \times 20 = 160 \quad (3)$$

distinct scenarios. Each is solved under both capacity conditions, producing 320 planner executions.

The scenarios are paired: package positions, deadlines, warehouse geometry, and all other initial conditions remain identical while only carrying capacity changes.

Scenario generation is fully automated. Each instance is associated with a scenario identifier, package count, layout mode, deadline mode, and reproducible random seed. Separate pseudo-random generators are used for layout and deadline generation. Therefore, changing the spatial layout does not unintentionally alter the deadline values associated with the same seed and package-count condition.

For every scenario, the generator creates two problem files that differ only in the initial value of `max-capacity`. This paired construction is important because performance differences can be attributed to carrying capacity rather than to differences between problem instances.

All problems are solved with ENHSP using the `opt-hrmax` configuration and a timeout of 120 seconds.

The collected metrics are:

- solved/unsolvable status;
- plan length;
- mission makespan;
- expanded nodes;
- evaluated states;
- planning time.

Solvability is analyzed with an exact McNemar test. Quantitative metrics are compared with the Wilcoxon signed-rank test using scenarios solved under both capacities.

Three ENHSP runs reported invalid negative planning-time measurements. These values were treated as missing. Two occurred among jointly solved scenarios, leaving 114 valid runtime pairs.

A. Experiment Automation and Reproducibility

The experimental pipeline is implemented in Python. It generates the PDDL+ problem files, invokes ENHSP, enforces the timeout, stores complete raw planner output, extracts the relevant metrics, and associates each run with its scenario metadata. Processed results are written to CSV files for subsequent statistical analysis.

The project also exposes the planner through a ROS2 node. Planning requests are sent through the topic `/warehouse_experiment/request`, while results are published on `/warehouse_experiment/result`. An interactive Jupyter notebook acts as a client, allowing a user to select a scenario and capacity and execute the same ENHSP pipeline used by the batch experiment. The notebook additionally reproduces the principal analyses and figures from the final dataset.

V. RESULTS

A. Solvability

Capacity one solved 116 of 160 scenarios (72.50%), while capacity two solved 150 (93.75%), an increase of 21.25 percentage points.

All 34 discordant pairs favored capacity two. The exact McNemar test produced $p = 1.16 \times 10^{-10}$.

Figure 1 shows that the effect depends strongly on the experimental condition.

TABLE II
PAIRED SOLVABILITY OUTCOMES

	Cap. 2 solved	Cap. 2 unsolved
Cap. 1 solved	116	0
Cap. 1 unsolved	34	10

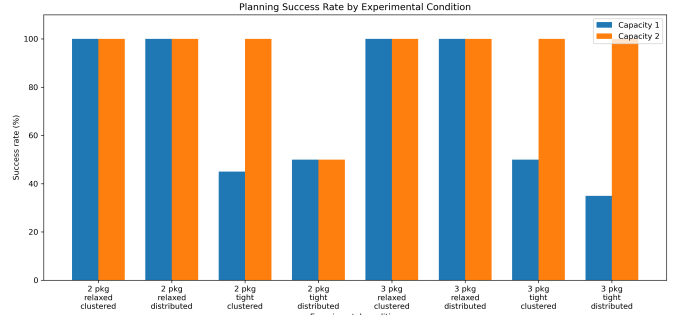


Fig. 1. Planning success rate by experimental condition.

Under relaxed deadlines, both capacities solved every scenario. Under tight deadlines, success changed from 45% to 100% for two clustered packages, 50% to 100% for three clustered packages, and 35% to 100% for three distributed packages. For two tight distributed packages, both capacities remained at 50%.

Table III reports all condition-level success rates.

The two-package tight distributed condition is especially informative. In this case, increasing capacity alone cannot remove the need to visit both aisles, and the observed feasibility rate is identical under both capacities.

B. Mission Efficiency

Among the 116 scenarios solved by both capacities, mean makespan decreased from 11.983 to 8.500 time units, corresponding to a 28.54% mean paired reduction. Capacity two produced lower makespan in 86 pairs and tied in 30; it was never worse. The Wilcoxon test produced $p = 4.89 \times 10^{-17}$.

Mean plan length decreased from 26.931 to 20.483 actions, a 23.34% mean paired reduction, with the same 86 improvements and 30 ties ($p = 4.89 \times 10^{-17}$).

For two-package distributed scenarios, mean makespan remained unchanged across capacities. In contrast, clustered scenarios produced large reductions because both packages could be collected during the same aisle visit.

C. Search Complexity

Capacity two also reduced planner search effort for most jointly solved scenarios.

Expanded nodes decreased by 61.93% on average ($p = 2.41 \times 10^{-20}$), while evaluated states decreased by 57.80% ($p = 2.93 \times 10^{-20}$). Capacity two required less search in 108 of 116 jointly solved scenarios.

Mean planning time decreased by 44.27% across the 114 valid runtime pairs ($p = 1.75 \times 10^{-18}$). Because runtime is

TABLE III
SUCCESS RATE BY EXPERIMENTAL CONDITION

Pkg.	Deadline	Layout	Cap. 1	Cap. 2
2	Relaxed	Clustered	100%	100%
2	Relaxed	Distributed	100%	100%
2	Tight	Clustered	45%	100%
2	Tight	Distributed	50%	50%
3	Relaxed	Clustered	100%	100%
3	Relaxed	Distributed	100%	100%
3	Tight	Clustered	50%	100%
3	Tight	Distributed	35%	100%

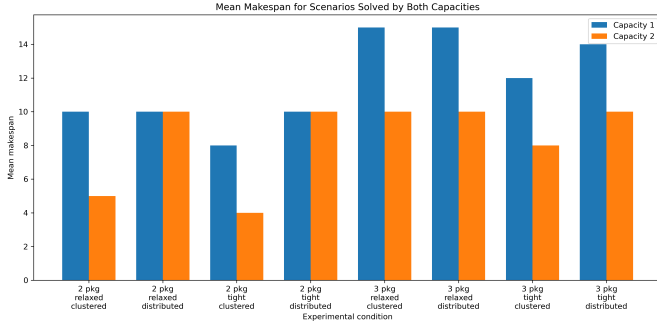


Fig. 2. Mean makespan for scenarios solved by both capacities.

sensitive to temporary system load, node and state counts are treated as the more robust measures of search effort.

D. ROS2 Pipeline Validation

The batch-planning pipeline is also exposed through ROS2 and an interactive Jupyter client. As an end-to-end consistency check, scenario 1000 was executed under both capacities. Capacity one returned plan length 20, makespan 8, 661 expanded nodes, and 1180 evaluated states; capacity two returned plan length 12, makespan 4, 50 expanded nodes, and 117 evaluated states. This confirms that the Jupyter-ROS2-ENHSP interface executes the same underlying planner workflow used by the offline experiment. The check is not counted as an additional statistical trial.

VI. DISCUSSION

The results show that additional carrying capacity can influence three different aspects of warehouse planning: feasibility, mission efficiency, and search complexity.

The spatial interaction is particularly important. When packages are clustered, a capacity-two robot can collect multiple packages during a single visit, reducing repeated travel. Conversely, when two packages are placed in different aisles, both aisles must be visited regardless of capacity. This condition therefore provides a useful contrast showing that the improvements cannot be attributed merely to changing a numeric model parameter.

The reduction in search effort is also notable. A larger capacity introduces additional possible carrying states and might therefore be expected to increase branching. In this

TABLE IV
PAIRED PERFORMANCE RESULTS

Metric	Cap. 1	Cap. 2	Reduction
Makespan	11.983	8.500	28.54%
Plan length	26.931	20.483	23.34%
Expanded nodes	12395.6	2544.4	61.93%
States evaluated	21328.6	5401.4	57.80%
Planning time (ms)	602.9	258.9	44.27%

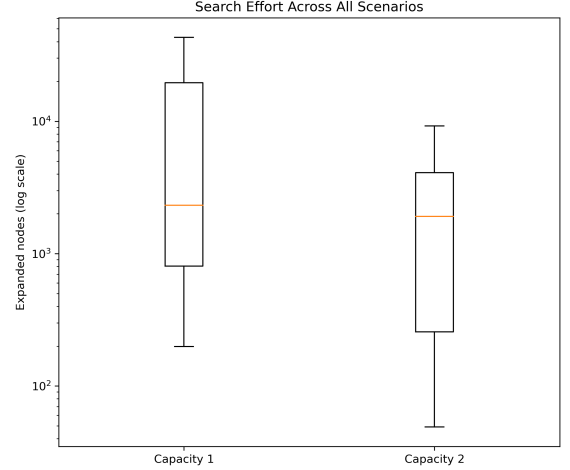


Fig. 3. Expanded-node distribution for the two capacity configurations.

domain, however, the planner generally reaches shorter solution strategies and explores fewer states. Capacity two still expanded more nodes in eight jointly solved scenarios, so the result should not be interpreted as a general theoretical property of higher-capacity planning.

VII. VALIDITY, LIMITATIONS, AND FUTURE WORK

a) *Construct validity.*: Carrying capacity is represented directly by the numeric fluent `max-capacity`, while `current-load` records the number of packages carried. Solvability, makespan, plan length, expanded nodes, evaluated states, and planning time capture complementary aspects of mission and planner performance. However, the synthetic deadlines, symbolic package handling, and small warehouse graph remain abstractions of an industrial system.

b) *Internal and statistical validity.*: The paired design holds package locations, deadlines, geometry, package count, layout, and random seed constant while changing only robot capacity. Separate deterministic random streams prevent layout changes from altering deadlines. Twenty seeds are evaluated per experimental condition. Solvability is analyzed with an exact McNemar test and paired quantitative outcomes with the Wilcoxon signed-rank test. Three invalid negative ENHSP timing values were treated as missing rather than corrected or imputed.

c) *External and algorithmic validity.*: The experiment uses one robot, two aisles, two or three packages, synthetic workloads, and a single ENHSP configuration (`opt-hrmax`).

The results therefore characterize this controlled domain rather than large industrial warehouses or all numeric planners. Four-package pilot instances showed a sharp increase in capacity-one search effort and were excluded from the balanced final experiment.

d) Future work.: Future experiments should consider larger warehouse graphs, additional package counts and capacity levels, multi-robot coordination, alternative planners and heuristics, repeated runtime trials, and realistic warehouse order traces.

VIII. CONCLUSION

This work evaluated how increasing robot carrying capacity from one to two packages affects deadline-constrained warehouse planning.

Across 160 paired scenarios, capacity two increased solvability from 72.50% to 93.75%. For the 116 scenarios solved by both capacities, it reduced mean makespan by 28.54%, plan length by 23.34%, expanded nodes by 61.93%, and evaluated states by 57.80%.

The results also demonstrate that these benefits depend strongly on package distribution. Additional capacity is most useful when several packages can share part of the same route, whereas spatially separated tasks may receive little or no route-level benefit.

Carrying capacity should therefore be considered jointly with task layout and temporal constraints when designing warehouse planning systems.

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